# APPLICATION OF ARTIFICIAL INTELLIGENCE METHODS IN THE REQUIREMENTS ENGINEERING MANAGEMENT PROCESS

Faculty of Fundamental Sciences – Department of Information Systems

**Supun Chandimal Thilakarathna Ranasinghe Mudiyanselage** | Supervisor: Assoc. Prof. Dr. Jolanta Miliauskaitė

## PROBLEM

Manual requirements engineering (RE) is slow, inconsistent and error-prone. Projects suffer from missed dependencies, undetected conflicts, and vague specifications that surface late in development — when fixing them is expensive. Up to 40% of project defects originate in the requirements phase.

## PROPOSED SOLUTION

A three-stage AI pipeline that automates core RE tasks while keeping a human in the loop at every low-confidence decision point.

### 01 Intelligent Structuring

LLM extracts actors, actions & constraints from raw text. Detects incomplete requirements and generates targeted clarification questions automatically.

### 02 Multi-Model Classification

Consensus voting ensemble — fine-tuned NoRBERT + Qwen2.5-7B LLM + rule-based classifier. Requires ≥2 classifiers to agree before accepting a classification.

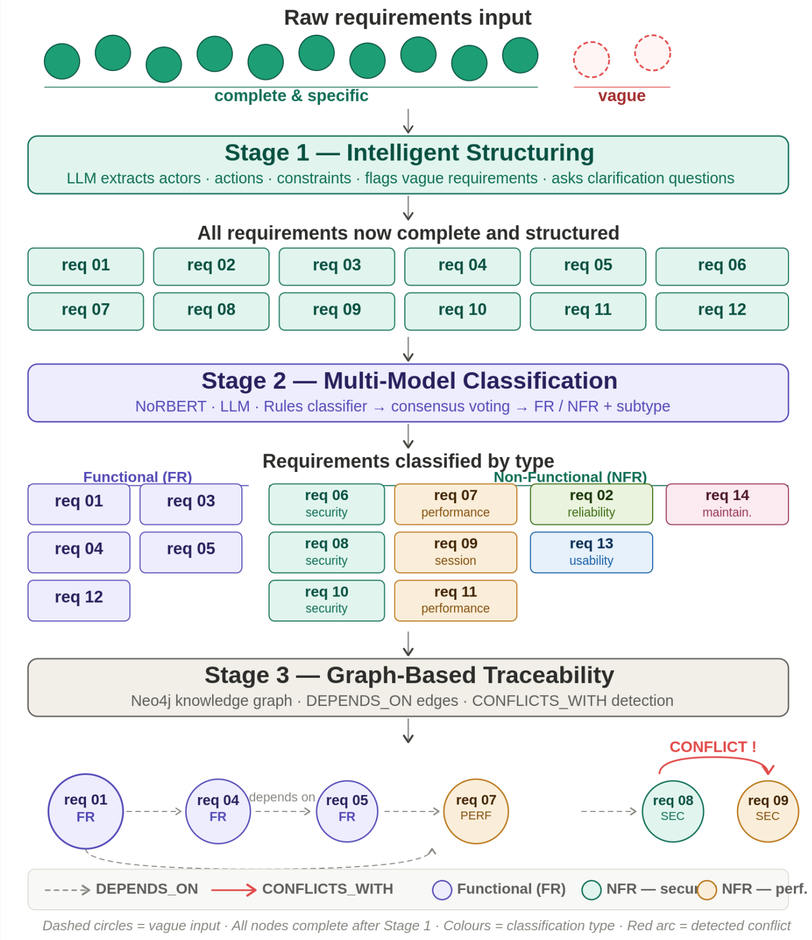
### 03 Graph-Based Traceability

Requirements stored as Neo4j knowledge graph nodes. Automatically detects DEPENDS_ON and CONFLICTS_WITH relationships via LLM reasoning, semantic similarity, and keyword matching.

## CONCLUSIONS

- Multi-model consensus voting eliminates hallucinated classifications — zero false positives across 125 independent test cases.
- Graph-based traceability automatically detected a critical semantic conflict (session timeout: 15 min vs 30 days) that manual review would likely have missed.
- End-to-end processing in 310.71 seconds — a 20–40× improvement over equivalent manual analysis.
- The ensemble's zero false positive guarantee is valuable in safety-critical RE contexts where false classifications misdirect architectural effort.
- Domain-specific fine-tuning (NoRBERT, F1=0.976) is the strongest single classifier — weighted voting is the clearest improvement pathway.

## SYSTEM ARCHITECTURE




## ANTI-HALLUCINATION DESIGN

### Consensus Voting Rules:

- All 3 agree → HIGH confidence → Auto-approve
- 2 of 3 agree → MEDIUM confidence → Majority accepted
- All disagree → LOW confidence → Manual review queue

*P(consensus hallucination) ≈ 0.001*

## EVALUATION DATASETS

**Banking Domain Batch (n=12)**

Custom requirements — FR, NFR, vague, and conflicting entries. Full end-to-end pipeline evaluation across all three stages.

**PROMISE NFR Benchmark (n=625)**

Pre-labelled requirements from real open-source projects. Last 125 held-out as independent test set for Stage 2 evaluation.

## FUTURE WORK

- Implement weighted voting to leverage NoRBERT's superior classification performance within the ensemble.
- Validate the pipeline across multiple application domains beyond banking — healthcare, e-commerce, and government systems.
- Fine-tune NoRBERT for multi-class NFR subtype prediction to improve Usability and Maintainability classification accuracy.
- Conduct a controlled user study measuring whether Stage 1 clarification questions are genuinely useful to practising RE engineers.

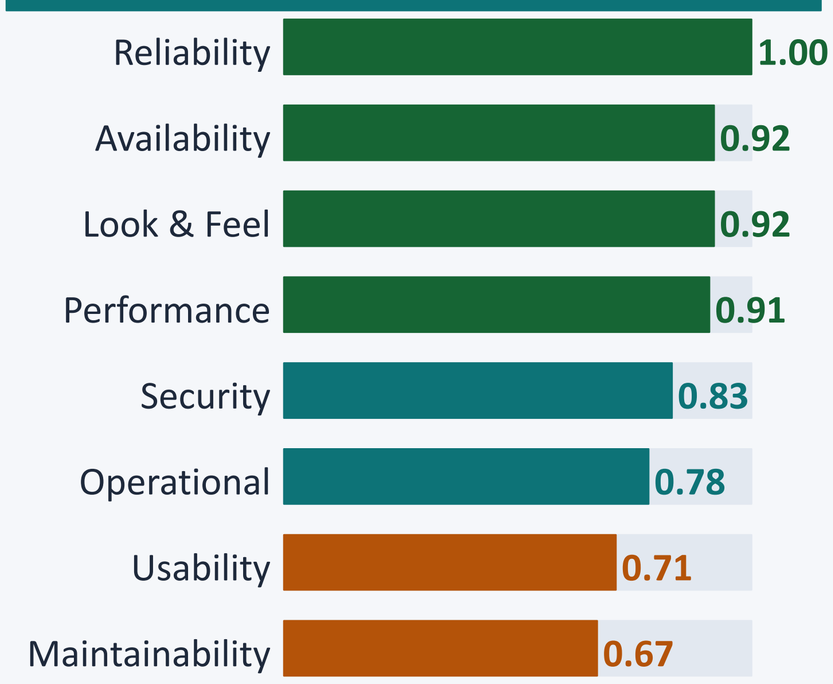
## KEY RESULTS

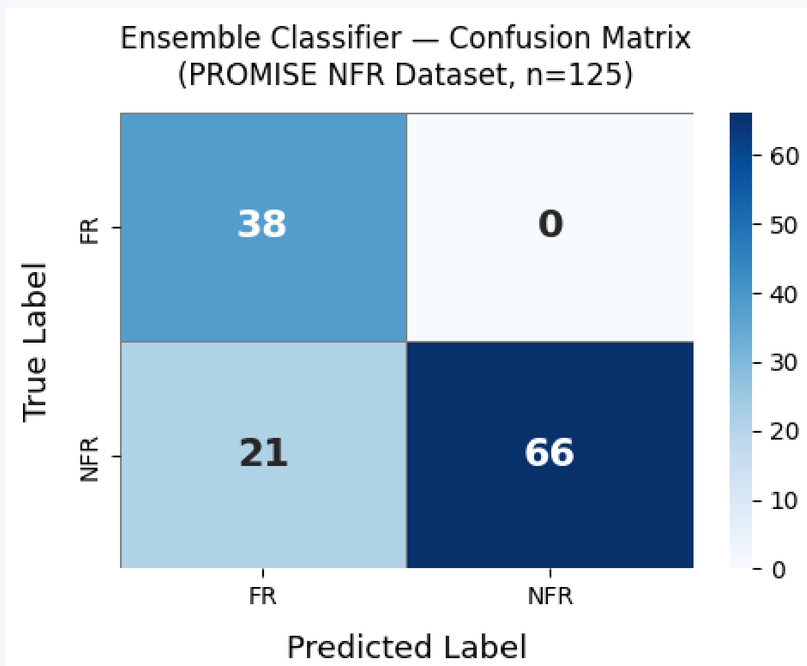
| 100% | 83.3% | 96.8% |
|---|---|---|
| Stage 1 Accuracy | Automation Rate | NoRBERT Accuracy |

## STAGE 2 CLASSIFICATION — PROMISE (N=125)

| Model | Accur. | Prec. | Recall | F1 |
|---|---|---|---|---|
| Rules | 63.2% | 84.7% | 57.5% | 0.685 |
| LLM | 66.4% | 100% | 51.7% | 0.682 |
| NoRBERT | **96.8%** | 100% | 95.4% | **0.976** |
| **Ensemble ★** | **83.2%** | **100%** | **75.9%** | **0.863** |

## NFR SUBTYPE F1 SCORES




## STAGE 3: TRACEABILITY RESULTS

| | |
|---|---|
| Dependencies detected: | **19** |
| Conflicts detected: | **1 (15 min vs 30 day session conflict)** |
| Orphan node rate: | **0%** |
| Graph density: | **0.30 (target: 0.15–0.30)** |
| Precision / Recall / F1: | **52.94% / 75.00% / 0.65** |